Exploratory Data Analysis: Car Fuel Efficiency

Introduction

Fuel efficiency is a critical factor in the automotive industry as it impacts consumer purchasing decisions, environmental sustainability, and regulatory compliance. Automobile manufacturers continuously aim to design vehicles that deliver better performance while maintaining optimal fuel consumption.

Dataset Overview

The dataset used in this analysis contains various attributes of cars such as miles per gallon (MPG), engine displacement, horsepower, number of cylinders, weight, acceleration, model year, and origin. These features provide insights into vehicle performance and fuel consumption characteristics.

Business Problem

Automobile manufacturers and market analysts want to understand which vehicle characteristics most strongly influence fuel efficiency. Identifying these factors can help companies design more fuel-efficient vehicles, reduce environmental impact, and meet changing market demands.

Objective of the Analysis

The objective of this exploratory data analysis (EDA) is to examine the dataset and identify relationships between fuel efficiency (MPG) and other vehicle attributes such as horsepower, weight, cylinders, and engine displacement. Through statistical exploration and visualization, this analysis aims to uncover patterns and correlations within the data.

Tools and Technologies

The analysis will be conducted using the following Python libraries:

- NumPy for numerical computations
- Pandas for data manipulation and analysis
- Matplotlib for data visualization
- Seaborn for statistical visualization

Expected Outcomes

Through this exploratory analysis, the goal is to:

- Identify key factors that influence vehicle fuel efficiency
- Understand relationships between engine characteristics and fuel consumption

- Detect patterns, correlations, and potential outliers in the dataset
- Generate insights that can support data-driven decision making in automotive design and performance optimization

Dataset Description

The dataset contains detailed trip-level fuel efficiency and performance data for different vehicles. Each row represents a specific trip taken by a vehicle, and the dataset captures multiple operational, efficiency, and environmental metrics.

Key variables in the dataset include:

- **trip_id** – Unique identifier for each trip.
- **car_name** – Name or model of the vehicle used for the trip.
- **vehicle_type** – Type of vehicle (e.g., sedan, SUV, hatchback).
- **fuel_type** – Type of fuel used by the vehicle.
- **trip_category** – Category of the trip such as city, highway, or mixed driving.
- **distance_km** – Total distance traveled during the trip in kilometers.
- **trip_duration_hr** – Duration of the trip in hours.
- **avg_speed_kmph** – Average speed during the trip.
- **base_mileage_kmpl** – Expected or benchmark fuel efficiency of the vehicle.
- **actual_mileage_kmpl** – Actual mileage achieved during the trip.
- **mileage_delta_kmpl** – Difference between expected and actual mileage.
- **fuel_consumed_L** – Total fuel consumed during the trip.
- **fuel_cost_usd** – Cost of fuel used for the trip.
- **cost_per_km_usd** – Fuel cost per kilometer.
- **co2_emissions_kg** – Total CO_2 emissions generated during the trip.
- **co2_per_km** – Emissions generated per kilometer traveled.
- **relative_efficiency** – Efficiency score comparing actual performance to expected performance.
- **efficiency_band / efficiency_class** – Categorization of the trip based on efficiency levels.
- **outlier_flag** – Indicates whether the trip contains abnormal or unusual efficiency values.
- **data_quality_score** – Quality score indicating the reliability of the recorded trip data.

```
import pandas as pd
import numpy as np
import seaborn as sns
import plotly.express as px
import matplotlib.pyplot as plt
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

import plotly.io as pio
pio.renderers.default = "notebook"
```

```
df = pd.read_csv(r"C:\Users\shubh\Shubham\EDA\Cars Fuel Efficiency_
Benchmark Edition -.csv")
```

Initial Data Exploration

Before performing deeper analysis, it is important to understand the structure and quality of the dataset. This step involves examining the dataset dimensions, inspecting column names, reviewing data types, and identifying missing or duplicate values. These checks help ensure that the data is reliable and suitable for further analysis.

```
df.shape
```

```
(600, 27)
```

```
df.head()
```

```
   trip_id       car_name vehicle_type fuel_type        trip_category
\
0        1       Lexus ES       Luxury    Diesel  Very Long (>400 km)

1        2       Lexus ES       Luxury    Petrol   Medium (50–200 km)

2        3        Audi A6       Luxury    Petrol   Medium (50–200 km)

3        4   Maruti Swift    Hatchback    Diesel    Long (200–400 km)

4        5  Hyundai Creta          SUV    Diesel    Long (200–400 km)


   distance_km  trip_duration_hr  avg_speed_kmph  base_mileage_kmpl  \
0       437.85              4.84           90.51              11.31
1       181.86              3.54           51.43               9.98
2       118.07              1.46           80.62              11.06
3       343.16              4.72           72.65              21.17
4       371.03              3.87           95.78              13.01

   actual_mileage_kmpl  ...  efficiency_band  fuel_efficiency_ratio  \
0                11.24  ...          Average               0.993811
1                 9.61  ...             Poor               0.962926
2                11.24  ...          Average               1.016275
3                19.65  ...          Average               0.928200
4                12.39  ...          Average               0.952344

   speed_efficiency_ratio  cost_to_emission_ratio
trip_efficiency_profile  \
0                8.052491                0.427054
Highway
1                5.351717                0.547535
Mixed
2                7.172598                0.547516
```

```
Highway
3                 3.697201                0.426987
Mixed
4                 7.730428                0.427080
Highway

   trip_intensity  efficiency_class benchmark_flag  outlier_flag  \
0       90.464876            Medium          False         False
1       51.372881               Low          False         False
2       80.869863            Medium          False         False
3       72.703390              High          False         False
4       95.873385            Medium          False         False

   data_quality_score
0                 100
1                 100
2                 100
3                 100
4                 100

[5 rows x 27 columns]
```

```
df.columns
```

```
Index(['trip_id', 'car_name', 'vehicle_type', 'fuel_type',
'trip_category',
       'distance_km', 'trip_duration_hr', 'avg_speed_kmph',
       'base_mileage_kmpl', 'actual_mileage_kmpl',
'mileage_delta_kmpl',
       'fuel_consumed_L', 'fuel_cost_usd', 'cost_per_km_usd',
       'co2_emissions_kg', 'co2_per_km', 'relative_efficiency',
       'efficiency_band', 'fuel_efficiency_ratio',
'speed_efficiency_ratio',
       'cost_to_emission_ratio', 'trip_efficiency_profile',
'trip_intensity',
       'efficiency_class', 'benchmark_flag', 'outlier_flag',
       'data_quality_score'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 27 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   trip_id                  600 non-null    int64
 1   car_name                 600 non-null    object
 2   vehicle_type             600 non-null    object
 3   fuel_type                600 non-null    object
```

```
 4   trip_category            600 non-null    object
 5   distance_km              600 non-null    float64
 6   trip_duration_hr         600 non-null    float64
 7   avg_speed_kmph           600 non-null    float64
 8   base_mileage_kmpl        600 non-null    float64
 9   actual_mileage_kmpl      600 non-null    float64
 10  mileage_delta_kmpl       600 non-null    float64
 11  fuel_consumed_L          600 non-null    float64
 12  fuel_cost_usd            600 non-null    float64
 13  cost_per_km_usd          600 non-null    float64
 14  co2_emissions_kg         600 non-null    float64
 15  co2_per_km               600 non-null    float64
 16  relative_efficiency      600 non-null    float64
 17  efficiency_band          600 non-null    object
 18  fuel_efficiency_ratio    600 non-null    float64
 19  speed_efficiency_ratio   600 non-null    float64
 20  cost_to_emission_ratio   600 non-null    float64
 21  trip_efficiency_profile  600 non-null    object
 22  trip_intensity           600 non-null    float64
 23  efficiency_class         600 non-null    object
 24  benchmark_flag           600 non-null    bool
 25  outlier_flag             600 non-null    bool
 26  data_quality_score       600 non-null    int64
dtypes: bool(2), float64(16), int64(2), object(7)
memory usage: 118.5+ KB
```

```
df.isnull().sum()
```

```
trip_id                     0
car_name                    0
vehicle_type                0
fuel_type                   0
trip_category               0
distance_km                 0
trip_duration_hr            0
avg_speed_kmph              0
base_mileage_kmpl           0
actual_mileage_kmpl         0
mileage_delta_kmpl          0
fuel_consumed_L             0
fuel_cost_usd               0
cost_per_km_usd             0
co2_emissions_kg            0
co2_per_km                  0
relative_efficiency         0
efficiency_band             0
fuel_efficiency_ratio       0
speed_efficiency_ratio      0
cost_to_emission_ratio      0
trip_efficiency_profile     0
```

```
trip_intensity             0
efficiency_class           0
benchmark_flag             0
outlier_flag               0
data_quality_score         0
dtype: int64
```

```
df.duplicated().sum()
```

```
0
```

```
df.describe()
```

```
          trip_id  distance_km  trip_duration_hr  avg_speed_kmph  \
count  600.000000   600.000000        600.000000      600.000000
mean   300.500000   260.697950          3.619317       79.283950
std    173.349358   137.129999          2.334950       22.789384
min      1.000000    20.110000          0.180000       40.070000
25%    150.750000   143.050000          1.840000       59.127500
50%    300.500000   264.790000          3.295000       78.725000
75%    450.250000   376.195000          4.910000       98.810000
max    600.000000   499.470000         11.760000      119.950000

       base_mileage_kmpl  actual_mileage_kmpl  mileage_delta_kmpl  \
count         600.000000           600.000000          600.000000
mean           15.903467            15.938833            0.035367
std             3.706881             3.866744            0.921230
min             8.020000             7.380000           -1.860000
25%            13.217500            12.960000           -0.680000
50%            15.300000            15.480000           -0.085000
75%            18.432500            18.572500            0.770000
max            24.990000            27.370000            2.380000

       fuel_consumed_L  fuel_cost_usd  cost_per_km_usd
co2_emissions_kg  \
count       600.000000     600.000000       600.000000
600.000000
mean         17.359800      20.970867         0.080490
41.260068
std          10.237931      12.535096         0.021750
25.279817
min           1.040000       1.190000         0.044000
2.400000
25%           8.480000      10.310000         0.064700
21.226000
50%          16.710000      20.015000         0.076750
37.976000
75%          24.750000      29.962500         0.092925
58.688750
max          54.150000      68.500000         0.171300
```

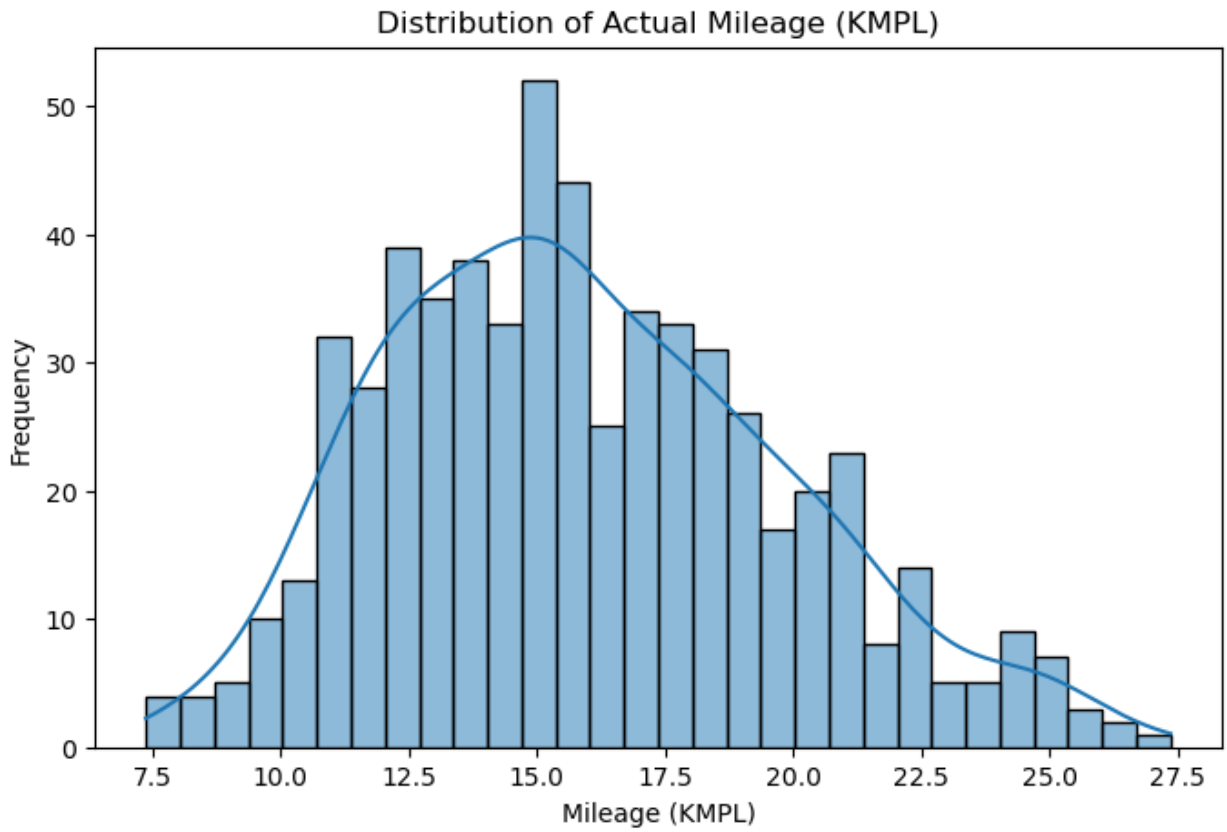
```
125.086000

       co2_per_km  relative_efficiency  fuel_efficiency_ratio  \
count  600.000000           600.000000             600.000000
mean     0.158652             0.709825               1.001559
std      0.045781             0.122779               0.056678
min      0.073100             0.392800               0.908672
25%      0.125025             0.621450               0.953670
50%      0.156700             0.703350               0.993616
75%      0.189800             0.791125               1.048907
max      0.312800             1.000000               1.109392

       speed_efficiency_ratio  cost_to_emission_ratio  trip_intensity
\
count              600.000000              600.000000      600.000000

mean                 5.297573                0.517176       79.270705

std                  2.117818                0.072331       22.779314

min                  1.907146                0.426466       40.027907

25%                  3.753873                0.427123       59.119145

50%                  4.946116                0.547607       78.760074

75%                  6.517981                0.601767       99.116183

max                 12.942242                0.604167      119.960714


       data_quality_score
count               600.0
mean                100.0
std                   0.0
min                 100.0
25%                 100.0
50%                 100.0
75%                 100.0
max                 100.0
```

Univariate Analysis

Univariate analysis helps understand the distribution of individual variables in the dataset. This step allows us to observe patterns, detect potential outliers, and understand the range and spread of important metrics such as fuel efficiency, fuel consumption, and emissions.

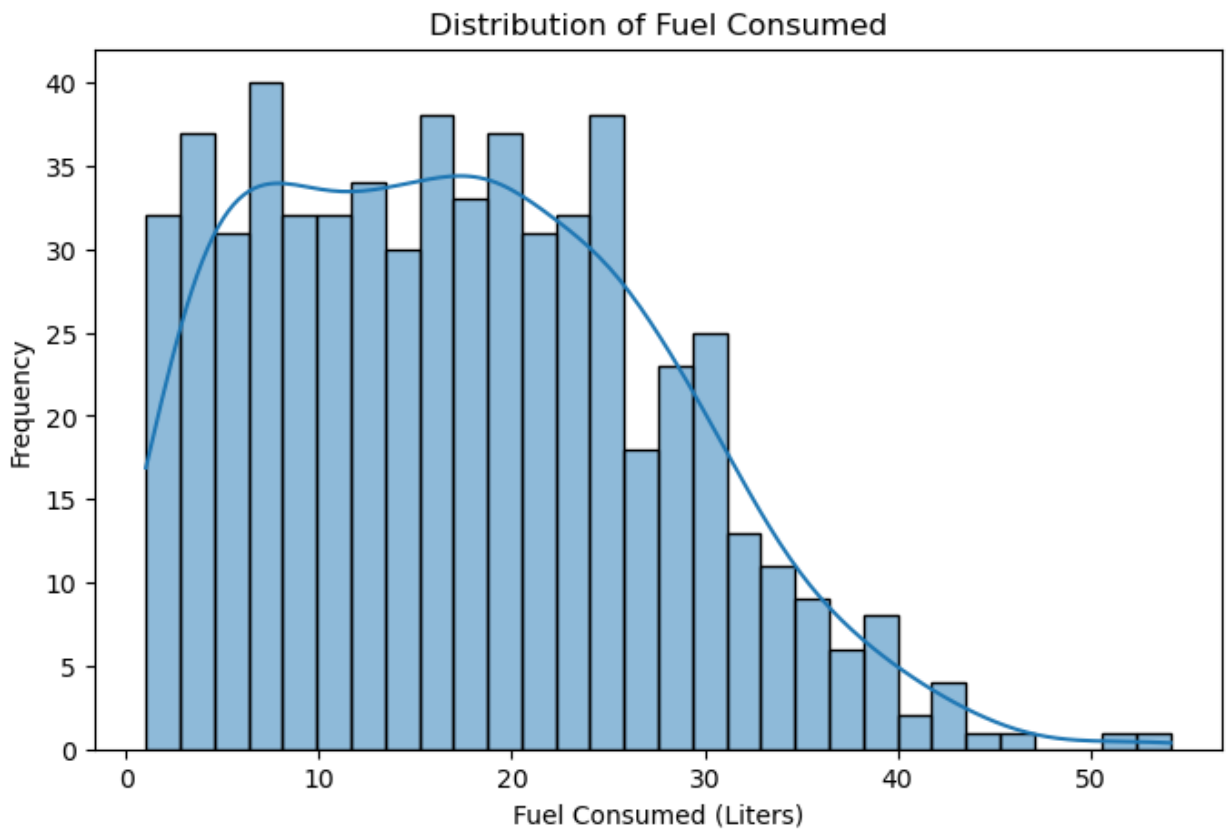
Distribution of Fuel Efficiency

```
plt.figure(figsize=(8,5))
sns.histplot(df['actual_mileage_kmpl'], bins=30, kde=True)
plt.title("Distribution of Actual Mileage (KMPL)")
plt.xlabel("Mileage (KMPL)")
plt.ylabel("Frequency")
plt.show()
```

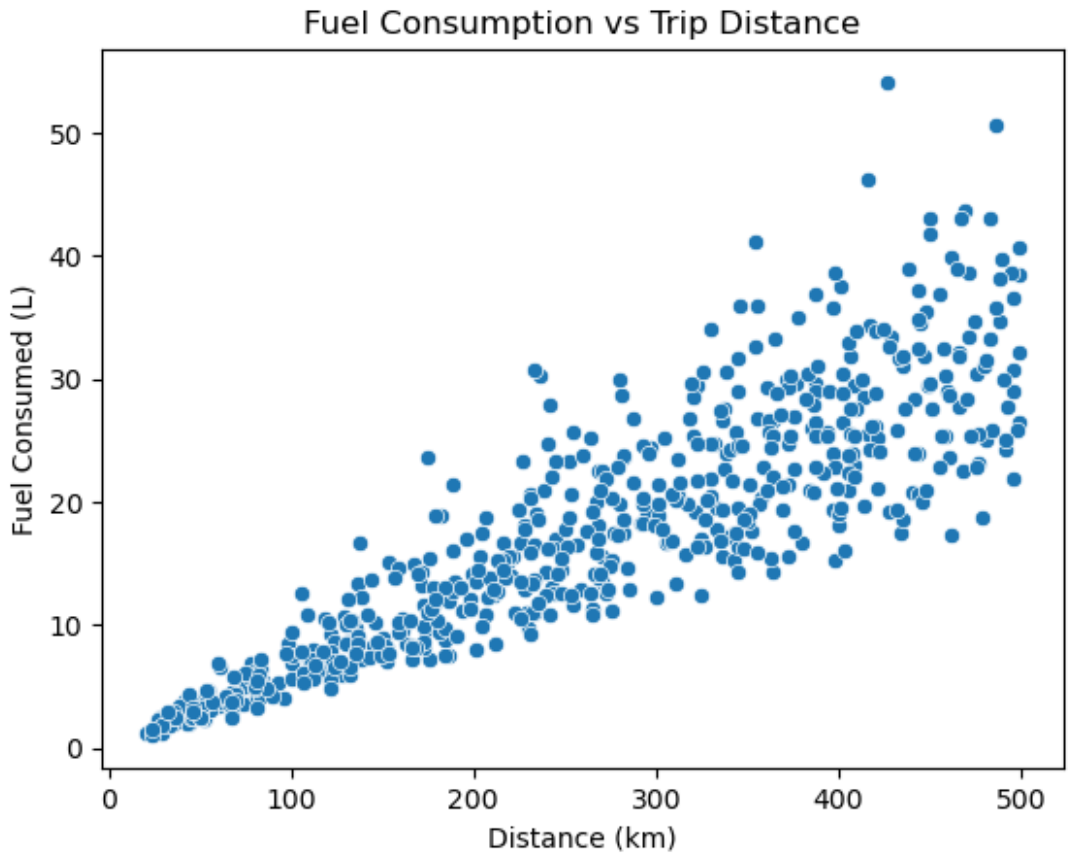


Distribution of Fuel Consumption

```
plt.figure(figsize=(8,5))
sns.histplot(df['fuel_consumed_L'], bins=30, kde=True)
plt.title("Distribution of Fuel Consumed")
plt.xlabel("Fuel Consumed (Liters)")
plt.ylabel("Frequency")
plt.show()
```

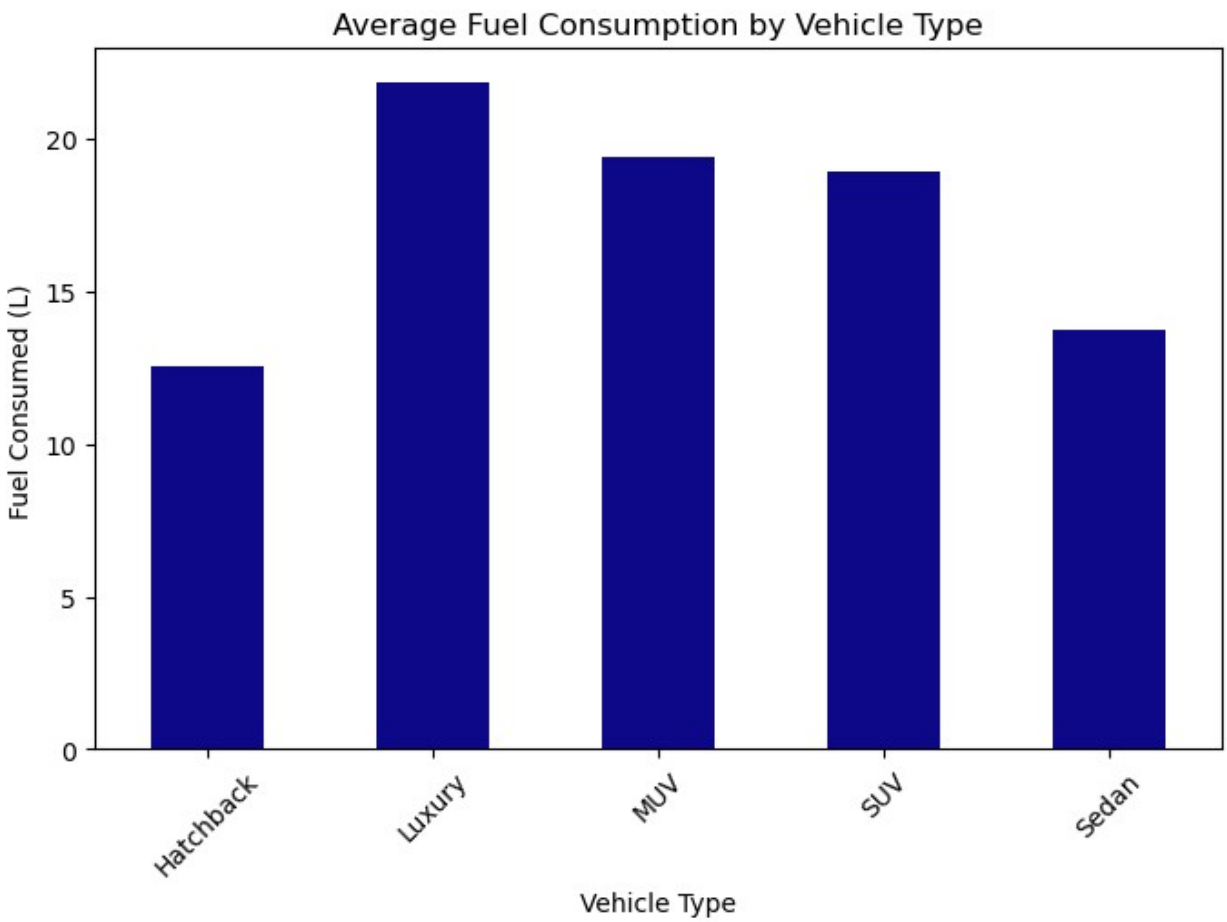



```
# Do longer trips consume more fuel?

sns.scatterplot(x='distance_km', y='fuel_consumed_L', data=df)
plt.title("Fuel Consumption vs Trip Distance")
plt.xlabel("Distance (km)")
plt.ylabel("Fuel Consumed (L)")
plt.show()
```



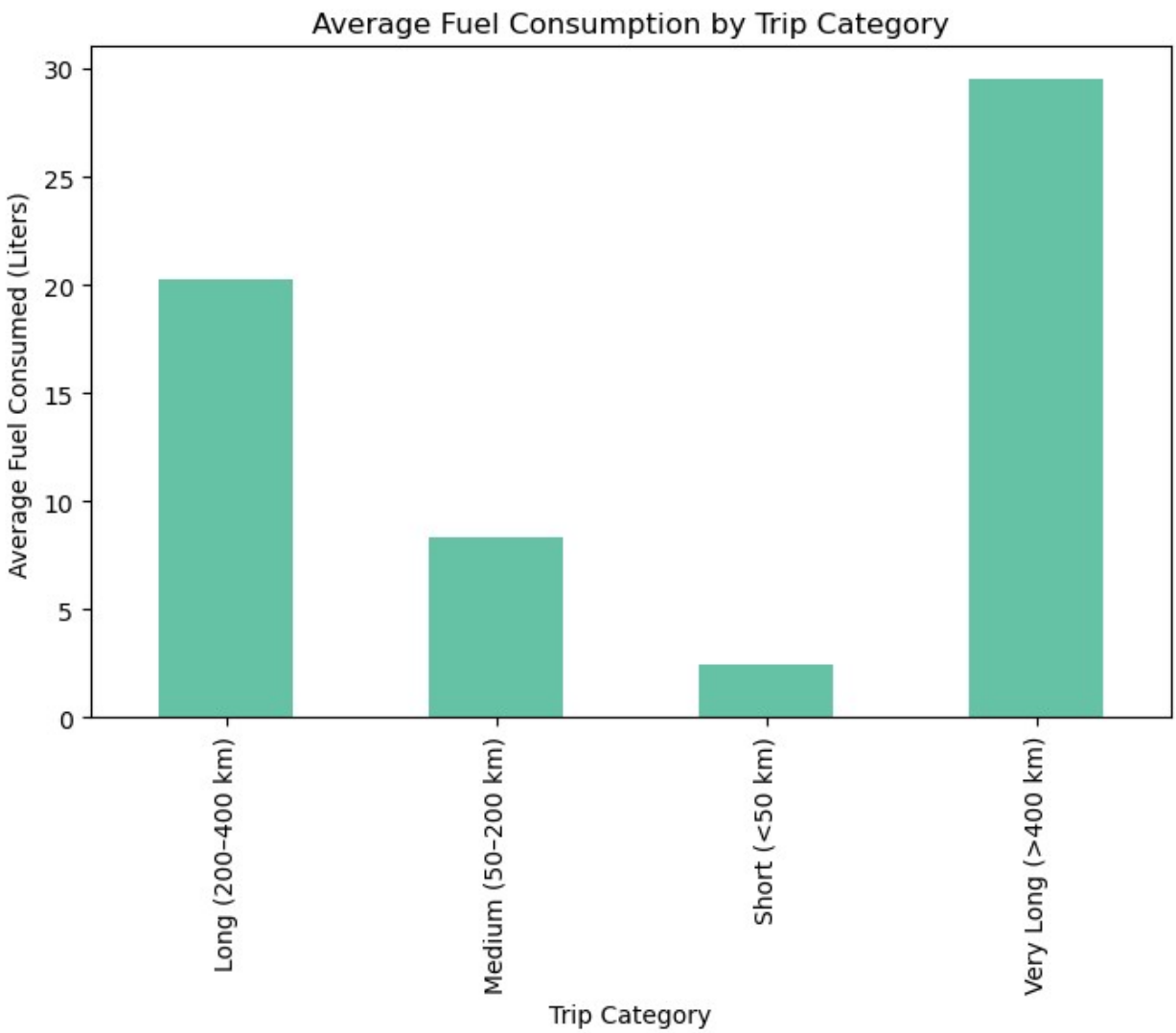
The scatter plot shows that fuel consumption increases as trip distance increases. This indicates that longer trips require more fuel, which aligns with expected vehicle behavior.

```
# Do different vehicle types consume different amounts of fuel?

plt.figure(figsize=(8,5))

df.groupby('vehicle_type')['fuel_consumed_L'].mean().plot(
    kind='bar',
    colormap='plasma'
)

plt.title("Average Fuel Consumption by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Fuel Consumed (L)")
plt.xticks(rotation=45)

plt.show()
```



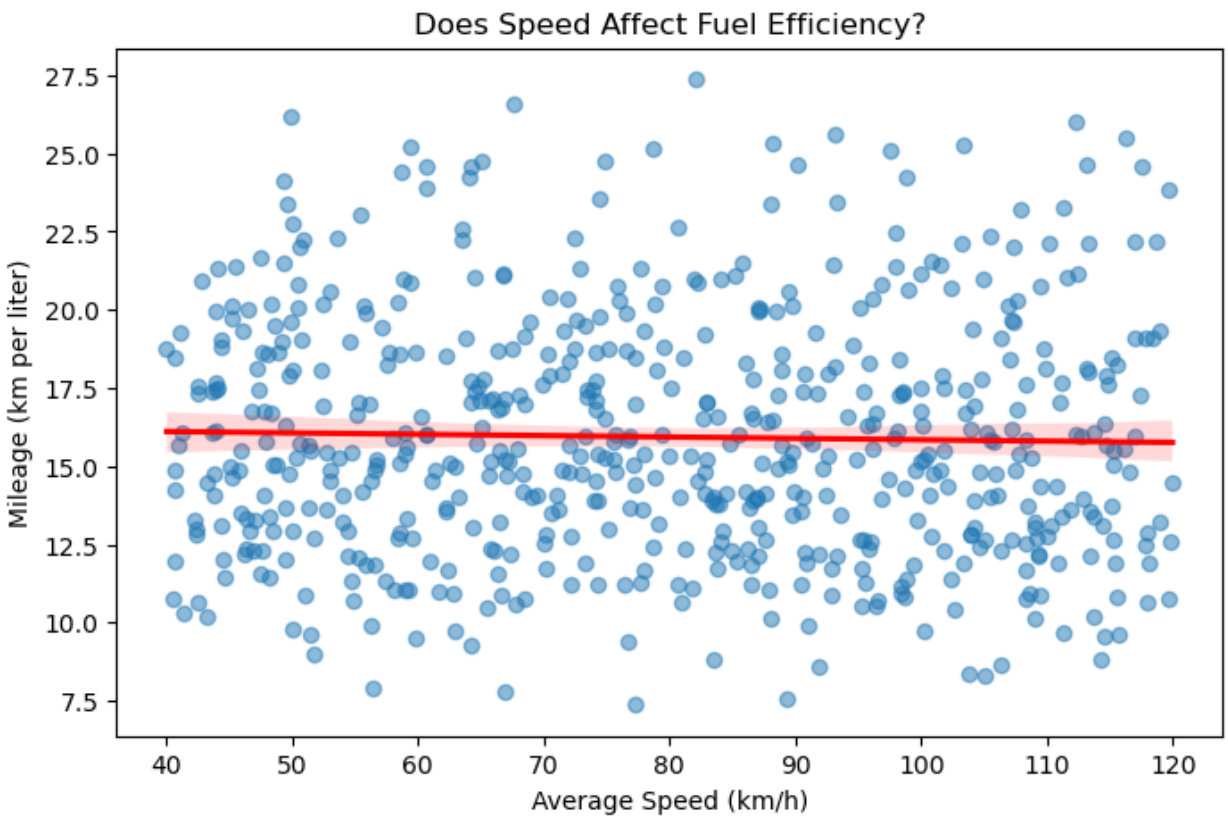
Which trip category consumes the most fuel?

```
plt.figure(figsize=(8,5))

df.groupby('trip_category')['fuel_consumed_L'].mean().plot(
    kind='bar',
    colormap='Set2'
)

plt.title("Average Fuel Consumption by Trip Category")
plt.xlabel("Trip Category")
plt.ylabel("Average Fuel Consumed (Liters)")

plt.show()
```



Do higher speeds reduce fuel efficiency?

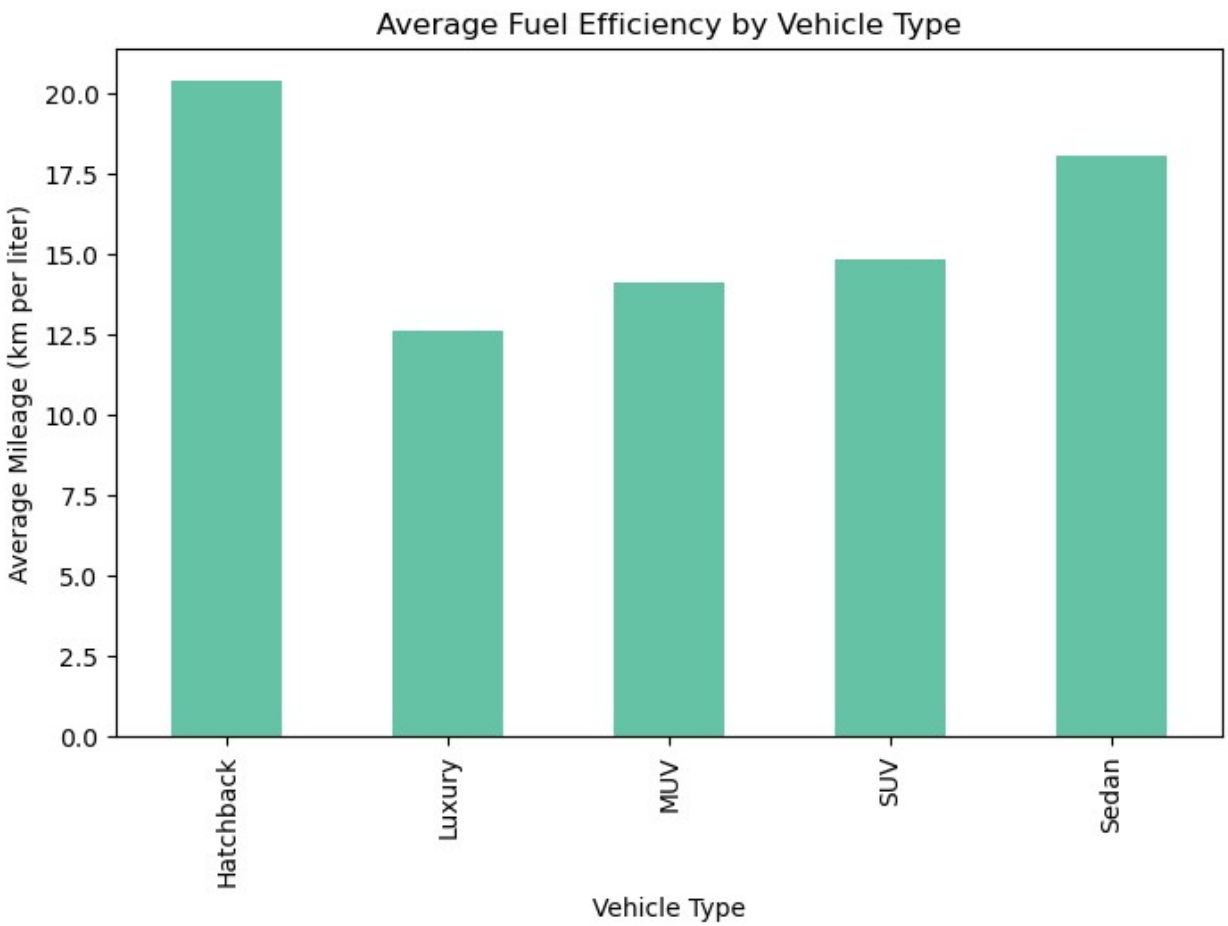
```
df['speed_range'] = pd.cut(df['avg_speed_kmph'], bins=5)

plt.figure(figsize=(8,5))

sns.regplot(x='avg_speed_kmph', y='actual_mileage_kmpl', data=df,
            scatter_kws={'alpha':0.5},
            line_kws={'color':'red'})

plt.title("Does Speed Affect Fuel Efficiency?")
plt.xlabel("Average Speed (km/h)")
plt.ylabel("Mileage (km per liter)")

plt.show()
```



The relationship between average speed and fuel efficiency appears weak. Mileage remains relatively stable across different speed levels, indicating that speed does not significantly affect fuel efficiency in this dataset.

Do different vehicle types have different fuel efficiency?

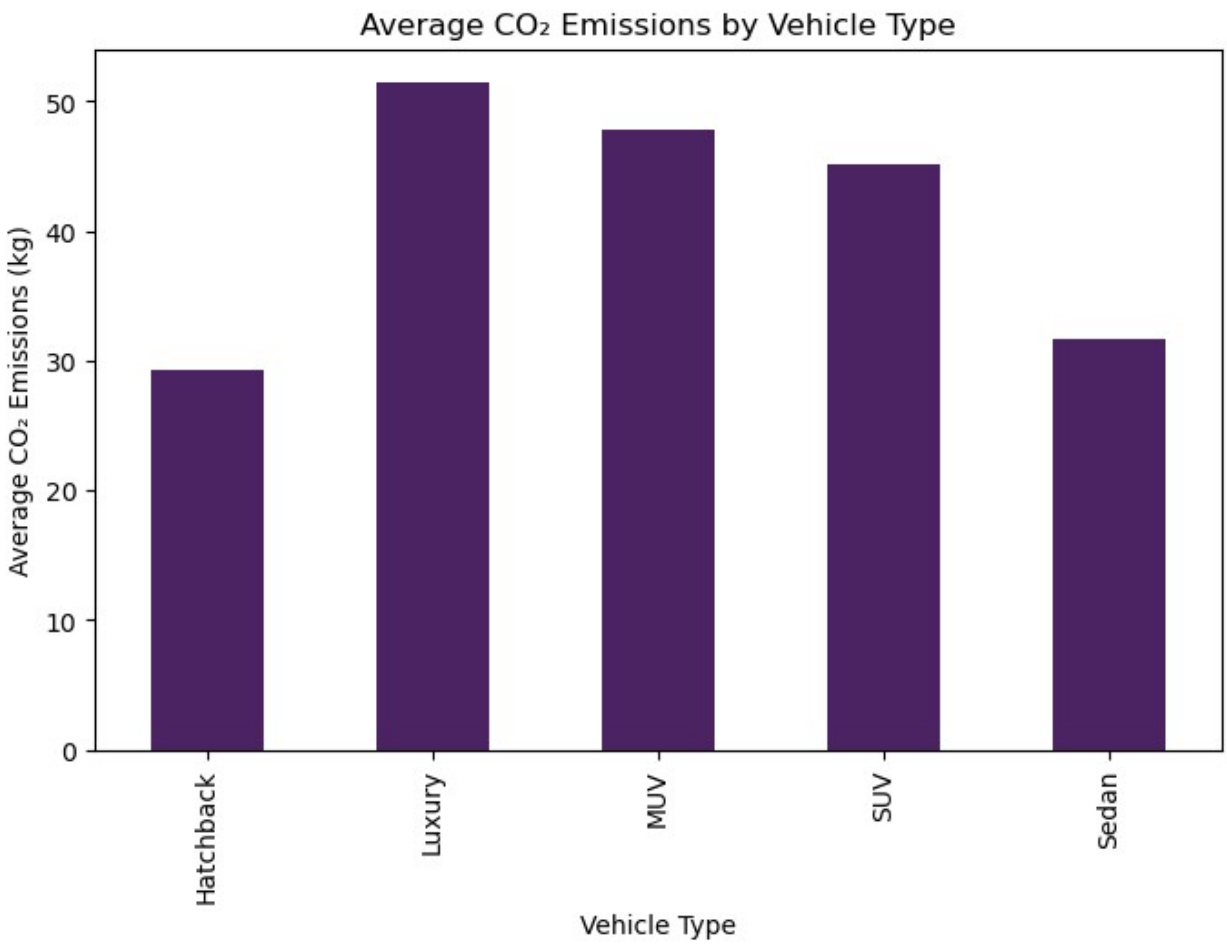
```
plt.figure(figsize=(8,5))

df.groupby('vehicle_type')['actual_mileage_kmpl'].mean().plot(
    kind='bar',
    colormap='Set2'
)

plt.title("Average Fuel Efficiency by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Average Mileage (km per liter)")

plt.show()
```



The chart shows that fuel efficiency varies across different vehicle types. Some vehicle types achieve higher average mileage compared to others, indicating that vehicle design and size influence fuel efficiency.

Which vehicle types produce the highest CO_2 emissions?

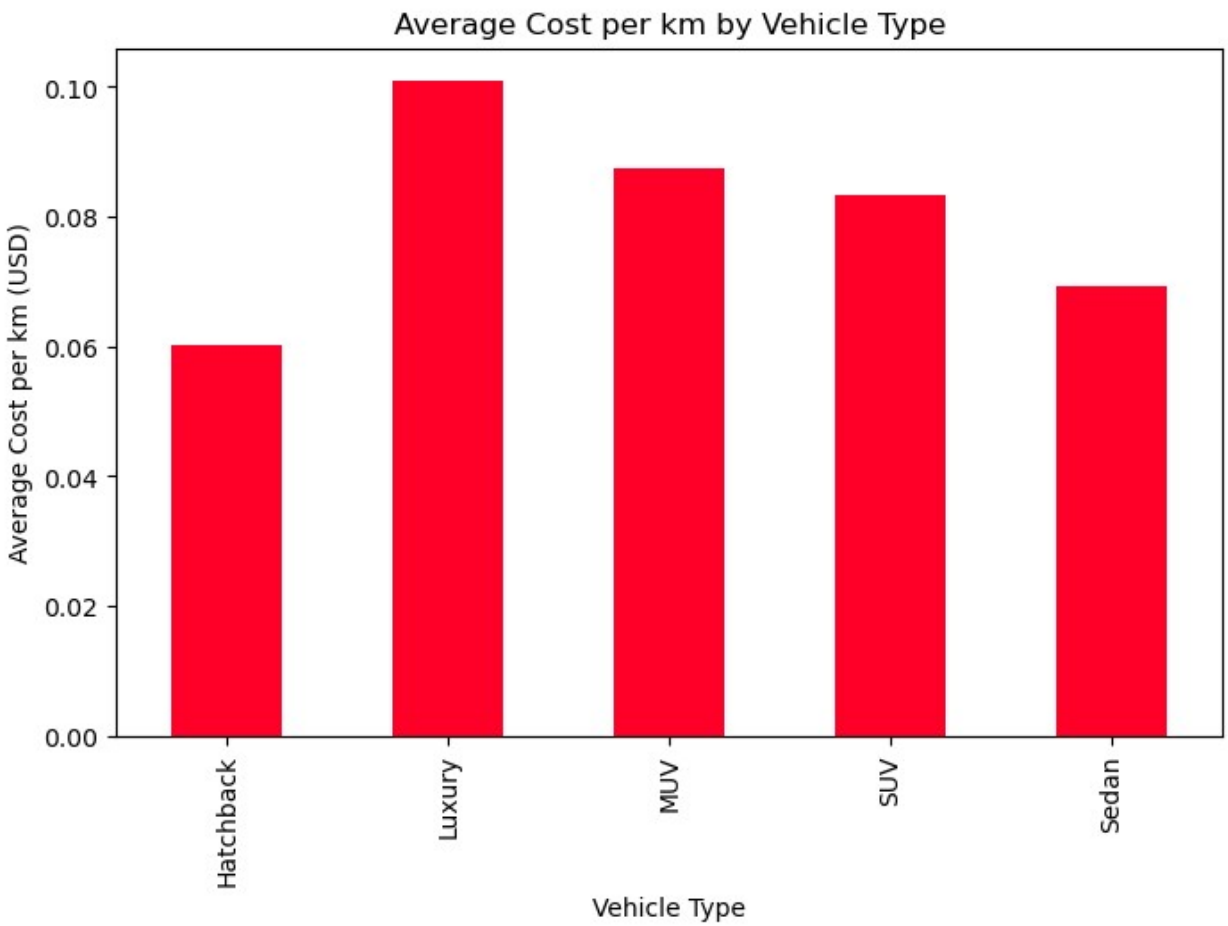
```
plt.figure(figsize=(8,5))

df.groupby('vehicle_type')['co2_emissions_kg'].mean().plot(
    kind='bar',
    colormap='flare_r'
)

plt.title("Average CO₂ Emissions by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Average CO₂ Emissions (kg)")

plt.show()
```



The chart shows that CO_2 emissions vary across different vehicle types. Some vehicle types generate higher average emissions per trip compared to others. This indicates that vehicle type has a significant impact on environmental impact and carbon emissions.

Which vehicle types are the most cost-efficient (lowest cost per km)?

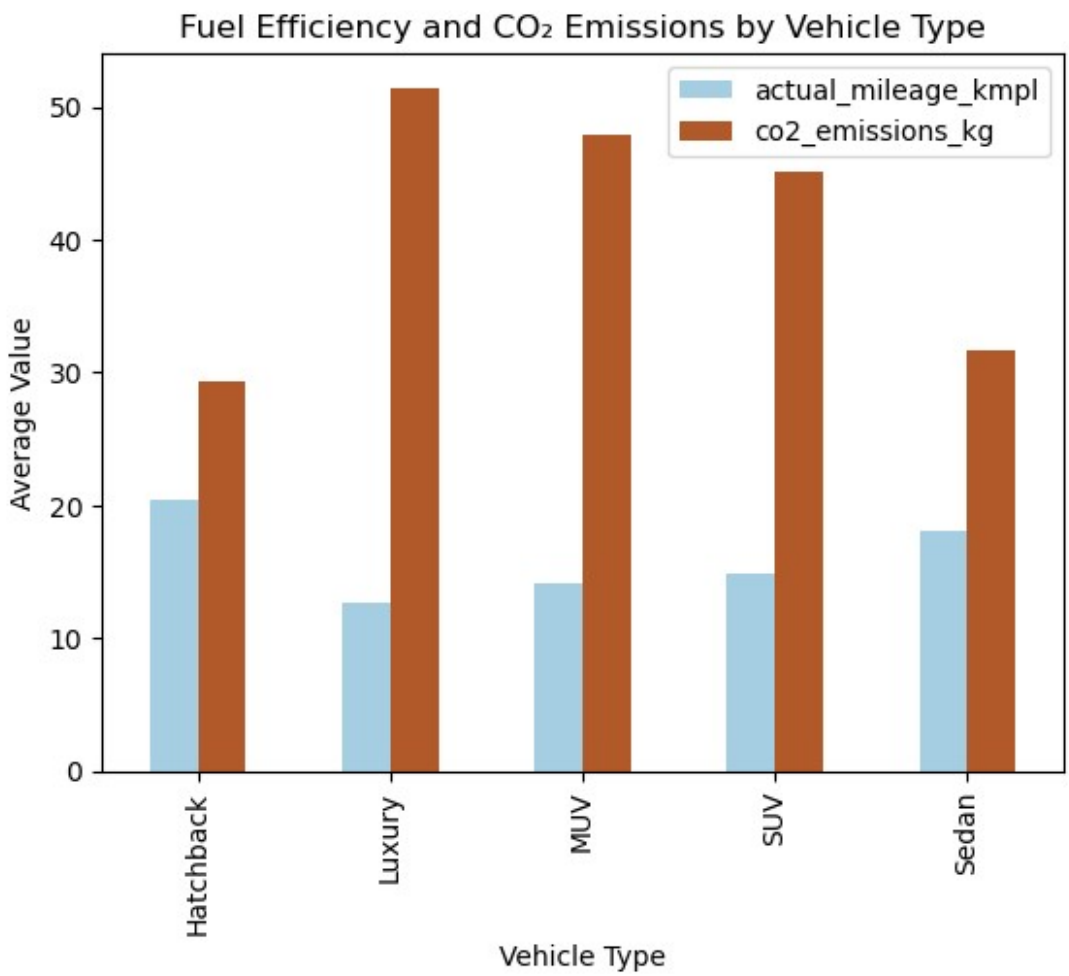
```
plt.figure(figsize=(8,5))

df.groupby('vehicle_type')['cost_per_km_usd'].mean().plot(
    kind='bar',
    colormap='gist_rainbow'
)

plt.title("Average Cost per km by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Average Cost per km (USD)")

plt.show()
```



Which vehicle types are both fuel-efficient and low-emission?

```
plt.figure(figsize=(8,5))

df.groupby('vehicle_type')
[['actual_mileage_kmpl','co2_emissions_kg']].mean().plot(
    kind='bar',
    colormap='Paired'
)

plt.title("Fuel Efficiency and CO₂ Emissions by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Average Value")

plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```



Which vehicle types have the highest fuel cost per trip?

```
import plotly.express as px

avg_cost = df.groupby('vehicle_type')
['fuel_cost_usd'].mean().reset_index()

fig = px.bar(
    avg_cost,
    x='vehicle_type',
    y='fuel_cost_usd',
    title="Average Fuel Cost per Trip by Vehicle Type",
    color='vehicle_type'
)

fig.show()
```

The chart shows that fuel cost per trip varies across vehicle types. Some vehicle types incur higher average fuel costs compared to others, indicating differences in fuel consumption and operational expenses.

Which trips generate the highest fuel cost per kilometer?

```
import plotly.express as px

vehicle_counts = df['vehicle_type'].value_counts().reset_index()
vehicle_counts.columns = ['vehicle_type', 'count']

fig = px.pie(
    vehicle_counts,
    names='vehicle_type',
    values='count',
    title="Distribution of Vehicle Types in Trips"
)

fig.show()
```

Exploring Machine Learning Possibilities

After understanding the dataset through exploratory data analysis, it is interesting to explore whether machine learning models could help predict certain outcomes such as fuel efficiency or fuel cost.

This section performs a simple experiment to see if trip characteristics such as distance and speed can be used to predict fuel efficiency.

Model Evaluation: R^2 Score

To evaluate the performance of the regression model, the R^2 (coefficient of determination) score is used.

The R^2 score measures how well the predicted values from the model match the actual values in the dataset. It indicates how much of the variation in the target variable can be explained by the input features.

The R^2 value ranges from:

- **1** → Perfect prediction
- **0** → The model performs no better than predicting the average value
- **Negative values** → The model performs worse than simply predicting the average

Higher R^2 values indicate better predictive performance.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = df[['distance_km','avg_speed_kmph']]
y = df['actual_mileage_kmpl']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

from sklearn.metrics import r2_score

r2_score(y_test, predictions)
```

```
-0.030201343877791942
```

The regression model produced a very low R^2 score, indicating that the selected features (distance and average speed) are not sufficient to accurately predict fuel efficiency. This suggests that additional variables such as vehicle type, fuel type, or trip characteristics may play a more important role in determining mileage.

Conclusion

This exploratory data analysis examined trip-level vehicle performance data to understand the factors influencing fuel efficiency, operational cost, and environmental impact.

The analysis revealed that vehicle characteristics and trip conditions play an important role in determining fuel consumption and emissions. Differences were observed across vehicle types and fuel categories, indicating that some vehicles are more efficient and environmentally friendly than others. Trip distance also showed a clear relationship with fuel consumption, as longer trips naturally require more fuel.

However, certain variables such as average driving speed did not show a strong influence on fuel efficiency in this dataset, suggesting that other factors may play a more significant role in determining mileage.

Overall, the findings highlight the importance of selecting fuel-efficient vehicles and optimizing trip characteristics to reduce operational costs and environmental impact. Organizations managing vehicle fleets can use such insights to make better decisions regarding vehicle selection, trip planning, and sustainability strategies.

This analysis also explored a simple machine learning experiment to investigate whether trip characteristics could be used to predict fuel efficiency. Although the initial model showed

limited predictive power, it demonstrates how machine learning techniques could be further applied with additional features and improved modeling approaches.